



Full Length Article

Allogeneic - Adult

The clinical value of anal swabs for microbial detection in allogeneic haematopoietic stem cell transplantation



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The intestinal microbiota plays critical roles in allogeneic hematopoietic stem cell transplantation (allo-HSCT). Rapid and effective microbial detection methods have important guiding value for the selection of intervention strategies for allo-HSCT recipients. We evaluated the application of the anal swab test before transplantation in allo-HSCT recipients. A total of 120 patients who underwent anal swab testing before allo-HSCT were retrospectively analyzed and divided into 3 groups: sterile (aseptic growth-negative), G⁺ (gram-positive bacterial colonization), and G⁻ (gram-negative bacterial colonization). On 16S rRNA sequencing, gram-negative bacteria predominated in the G⁻ group before and after transplantation. Compared with the sterile group, the percentage of natural killer cells was higher and the percentage of T cells was lower after transplantation in the G⁻ group at 1 month after transplantation. The percentage of CD4⁺ and CD4⁺CD8⁺ T cells was lower and the percentage of regulatory T cells was higher in the G⁻ group. The plasma levels of proinflammatory cytokines (TNF- α , IFN- γ , IL-6, and IL-17A) at 2 weeks post-transplantation were lower in the G⁻ group than in the sterile group, as was the cumulative incidence of grade III-IV acute graft-versus-host disease (GVHD). Gram-negative bacterial colonization before allo-HSCT was associated with low rates of bloodstream infections within 100 days post-transplantation and cytomegalovirus reactivation at 100 days to 2 years post-transplantation. Moreover, patients in the G⁻ group had a higher rate of 2-year GVHD-free, relapse-free survival compared with patients in the sterile group. The detection results using anal swabs were consistent with the gram-negative or gram-positive bacteria abundance of 16S rRNA sequencing results and associated with immune homeostasis and clinical outcomes after allo-HSCT. Anal swab testing may have potential advantages as a simple and effective method for microbial detection in allo-HSCT.

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INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an effective treatment for a variety of hematologic diseases, but high-dose radiation, chemotherapy pretreatment, and immunosuppressant use may result in bone marrow suppression, immune decline, and an impaired mucosal barrier in patients undergoing allo-HSCT. Infection and acute graft-versus-host disease (aGVHD) are the major causes of death after allo-HSCT [1]. The intestinal microbiota plays important roles in regulating intestinal homeostasis and immune response. In allo-HSCT recipients, conditioning regimens lead to dysbiosis of the intestinal microbiota [2], which is associated

with various complications, including infection [3], aGVHD [4], and primary disease relapse [5].

Recent studies on gut microbiology in allo-HSCT recipients have relied mostly on fecal 16S amplicon sequencing and macrogenome sequencing technologies; however, these approaches are not time-sensitive for pathogen discovery and intervention guidance in clinical transplant recipients [6]. Indeed, rapid and accurate identification of pathogens in the pretransplantation period and the formulation of reasonable treatment plans in conjunction with clinical practice can effectively control the occurrence and development of post-transplantation-related complications, reduce medical costs, and improve prognosis [7]. The clinical anal swab test, a simple and rapid method for detecting lower gastrointestinal pathogens, provides relatively timely and effective guidance in the detection and treatment of pathogenic bacteria in

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infected patients [8]. However, there are no reports on the effectiveness of anal swab detection in allo-HSCT recipients and the correlation between anal swab detection and clinical progression.

In the present study, we conducted a retrospective study of 120 patients undergoing allo-HSCT at our institution from January 1, 2019, to December 31, 2020. We first analyzed the fecal microbiological characteristics of patients detected using anal swabs before allo-HSCT and then analyzed the immune reconstruction and prognosis of patients with different levels of microbiological colonization. This study reveals the potential advantages of anal swab detection before transplantation in clinical allo-HSCT recipients.

METHODS

Study Subjects

We retrospectively analyzed the data of 124 allo-HSCT recipients who underwent an anal swab test before or on the day of transplantation at the First Affiliated Hospital of Soochow University between January 1, 2019, and December 31, 2020. In a prospective clinical study (2020-007), 62 allo-HSCT recipients were enrolled for intestinal microbiota analysis at our center between January 1, 2019, and December 31, 2020. Fecal samples from these 62 patients were collected before and after transplantation for 16S rRNA sequencing analysis. We expanded the number of allo-HSCT patients to 124 in the same period at a ratio of 1:1 in this study. Among these 124 patients, 4 patients with both gram-negative and gram-positive bacteria of anal swab detection were excluded because of the small number of such cases, and 120 patients were ultimately included in this study. The study was approved by the Ethics Committee of the First Affiliated Hospital of Soochow University.

Characteristics of the allo-HSCT patients were recorded, including age, sex, initial diagnosis, time of transplantation, type of transplantation, pretreatment protocol, stem cell source, use of antithymocyte globulin, infused mononuclear and CD34⁺ cell counts, and use of antibiotics before transplantation. The treatment of patients was provided according to local protocols.

Anal Swab Test

One hundred twenty-four patients had fully documented anal swab test results from either before or on the day of transplantation. A sterile anal swab was inserted under aseptic conditions 2 to 3 cm into the anus, wiped in the folds around the anus or gently rotated inside the anus and then inserted into a sterile test tube for routine microbial culture. The main target microorganisms for identification and drug sensitivity testing were gram-positive bacteria (*Enterococcus faecalis*, *Staphylococcus epidermidis*, acid-fast bacilli, and *Bacillus cereus*) and gram-negative bacteria (*Escherichia coli*, *Klebsiella oxytoca*, and *Proteus mirabilis*). The distribution of patients with anal-swab cultured microorganisms is shown in Supplementary Table S1. Four patients with gram-positive bacteria and gram-negative bacteria simultaneously cultured from anal swabs were not included in this study.

One hundred twenty patients were divided into 3 groups based on the anal swab results: sterile group (aseptic growth-negative group), G⁺ group (gram-positive bacterial colonization group), and G⁻ group (gram-negative bacterial colonization group). Patients with a cultured gram-positive microorganism on the anal swab test were classified as the G⁺ group, and those with a cultured gram-negative microorganism composed the G⁻ group.

Fecal 16S rRNA Gene Sequencing

Fecal samples were collected before transplantation and at 7 to 14 days and 14 to 28 days after transplantation. Total genomic DNA was extracted, and DNA concentration and integrity were verified using a NanoDrop spectrophotometer (Thermo Fisher Scientific, Waltham, MA) and agarose gel electrophoresis. For bacterial diversity analysis, the V3 to V4 hypervariable regions of the 16S rRNA gene were amplified with universal primers (forward, 5'-TACGGRAGGCAGCAG-3'; reverse, 5'-AGGGTATCTAATCCT-3'). The 16S rRNA gene amplicon sequencing and analysis were performed by OE Biotechnology (Shanghai, China). The top 15 microbial genera in the sequencing results were analyzed in detail.

Flow Cytometry

Peripheral blood specimens were collected in EDTA anticoagulant tubes at 2 weeks, 1 month, 2 months, and 3 months after transplantation and processed within 1 hour for immune cell and cytokine analysis. The percentages of lymphocytes and different lymphocyte subsets were detected using flow cytometry. The samples were probed with the following antibodies: anti-CD45, anti-CD3, anti-CD19, anti-CD16, anti-CD56, anti-CD4, anti-CD8, anti-CD25 and anti-CD127 (Beckman Coulter, Brea, CA). Flow cytometry analysis was performed with a Navios flow cytometer (Beckman Coulter).

Peripheral blood specimens were centrifuged to collect plasma for cytokine analysis. The levels of TNF- α , IFN- γ , IL-6, IL-17A, IL-10, and IL-4 were determined by flow cytometry using corresponding Cytometric Bead Array reagents (BD Biosciences, San Diego, CA) and the BD FACSCalibur system (BD Biosciences).

Definitions

aGVHD was defined and graded from 0 to IV depending on the pattern and severity of organ damage according to established criteria [9]. Infection was assessed according to the Common Terminology Criteria for Adverse Events version 5.0 [10]. Cytomegalovirus (CMV) or Epstein-Barr virus (EBV) reactivation was defined as viral DNA load of >100 copies/mL on at least 1 measurement at any time after HSCT. Bloodstream infection (BSI) was defined as the isolation of bacteria from at least 1 blood culture. Survival probability refers to patients who survived at 2 years after allo-HSCT. Nonrelapse mortality (NRM) was defined as death without disease progression or relapse after HSCT. Overall survival (OS) was defined as the time from transplantation to death or last follow-up. Disease-free survival (DFS) was defined as survival without relapse. GVHD-free, relapse-free survival (GRFS) was defined as survival without grade III-IV aGVHD, extensive chronic GVHD, or relapse. Relapse was defined by morphologic evidence of disease in peripheral blood, bone marrow, or extramedullary sites.

Statistical Analysis

Categorical variables were compared using the chi-square test or Fisher exact test, and continuous variables were compared using the nonparametric Mann-Whitney *U* test. For data with skewed distributions, continuous variables are expressed as median with interquartile range (IQR). OS, DFS, and GRFS were estimated using the Kaplan-Meier method and the log-rank test. Competing risk analysis was used to calculate the cumulative incidences of aGVHD, infection, BSI, CMV, and EBV. Death from any cause was included as a competing risk for aGVHD, infection, BSI, and reactivation of CMV or EBV. The Fine-Gray test and log-rank test were applied for univariate

analysis. Cox proportional risk models and competing risk models were used in multivariate analyses to calculate hazard ratios (HRs) and 95% confidence intervals (CIs) and to identify independent variables associated with clinical endpoints. A P value $<.05$ was considered statistically significant. Multivariate Cox proportional risk regression models and survival analyses were conducted using SPSS software (IBM, Armonk, NY). Cumulative incidence rates were calculated using R version 4.2.2 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Patient Clinical Characteristics

A total of 120 patients who performed an anal swab test before or on the day of transplantation were retrospectively evaluated in this study. Based on the results of the anal swabs, the patients were classified into 3 groups: sterile group ($n = 39$), G^+ group ($n = 44$), and G^- group ($n = 37$) (Supplementary Table S1). Patient baseline information and clinical

characteristics are presented in Table 1. There were no significant differences among the 3 groups in terms of sex, age, diagnosis of primary disease, donor type, conditioning regimen, stem cell source, use of antithymocyte globulin, dose of infused mononuclear and $CD34^+$ cells, or use of antibiotics.

Intestinal Microbiota Characteristics

To determine whether the results of anal swab tests have diagnostic implications for the intestinal microbiota, we conducted 16S rRNA sequencing of stool specimens. The Simpson index showed no difference in the α -diversity of stool samples among the 3 groups (Figure 1A). Principal coordinate showed no differences in the β -diversity among the 3 groups before transplantation ($PP = .183$) and 7 to 14 days after transplantation ($P = .11$) (Figure 1B) but significant differences at 14 to 28 days after transplantation ($P = .004$) (Figure 1B). The relative abundances of gram-negative bacteria in the G^- group were higher than those in the sterile group before and after allo-HSCT (before allo-HSCT, 88.7% versus 70.9% [$P = .049$]; at 7 to

Table 1
Baseline Patient, Disease and Transplantation Characteristics

Variables	Sterile Group (N = 39)	G^+ Bacterial Colonization Group (N = 44)	G^- Bacterial Colonization Group (N = 37)	P Value
Sex, n (%)				.727
Female	16 (13.3)	17 (14.2)	12 (10.0)	
Male	23 (19.2)	27 (22.5)	25 (20.8)	
Age, yr				.849
≤ 40	20 (16.7)	21 (17.5)	20 (16.7)	
> 40	19 (15.8)	23 (19.2)	17 (14.2)	
Primary disease, n (%)				.23
AML	15 (16.7)	16 (16.7)	18 (16.7)	
ALL	8 (6.7)	18 (15)	11 (9.2)	
MDS	8 (6.7)	3 (2.5)	3 (2.5)	
Other	8 (6.7)	7 (5.8)	5 (4.2)	
Donor type, n (%)				.083
Mismatched	31 (25.8)	27 (22.5)	21 (17.5)	
Matched	8 (6.7)	17 (14.2)	16 (13.3)	
Conditioning, n (%)				.628
RIC	37 (30.8)	41 (34.2)	33 (27.5)	
MAC	2 (1.7)	3 (2.5)	4 (3.3)	
Stem cell source, n (%)				.59
BM	1 (0.8)	2 (1.7)	3 (2.5)	
BM/PB	11 (9.2)	17 (14.2)	14 (11.7)	
PB	27 (22.5)	25 (20.8)	20 (16.7)	
Antithymocyte globulin, n (%)				.248
No	22 (18.3)	32 (26.7)	26 (21.7)	
Yes	17 (14.2)	12 (10)	11 (9.2)	
Mononuclear cells, $\times 10^8/\text{kg}$				.557
Median	10.49	10.68	9.77	
IQR	(7.73-14.35)	(8.98-12.87)	(6.40-14.22)	
$CD34^+$ cells, $\times 10^6/\text{kg}$				.415
Median	3.76	4.225	4	
IQR	(2.64-5.25)	(3.20-5.93)	(3.14-5.54)	
Antibiotics, n (%)				.205
β -lactams	20 (16.7)	33 (27.5)	26 (21.7)	
Tetracyclics	5 (4.2)	2 (1.7)	0 (0)	
Quinolones	7 (5.8)	5 (4.2)	3 (2.5)	
Others	7 (5.8)	4 (3.3)	8 (6.7)	

AML indicates acute myelogenous leukemia; ALL, acute lymphoblastic leukemia; MDS, myelodysplastic syndrome; RIC, reduced-intensity conditioning; MAC, myeloablative conditioning; BM, bone marrow; PB, peripheral blood.

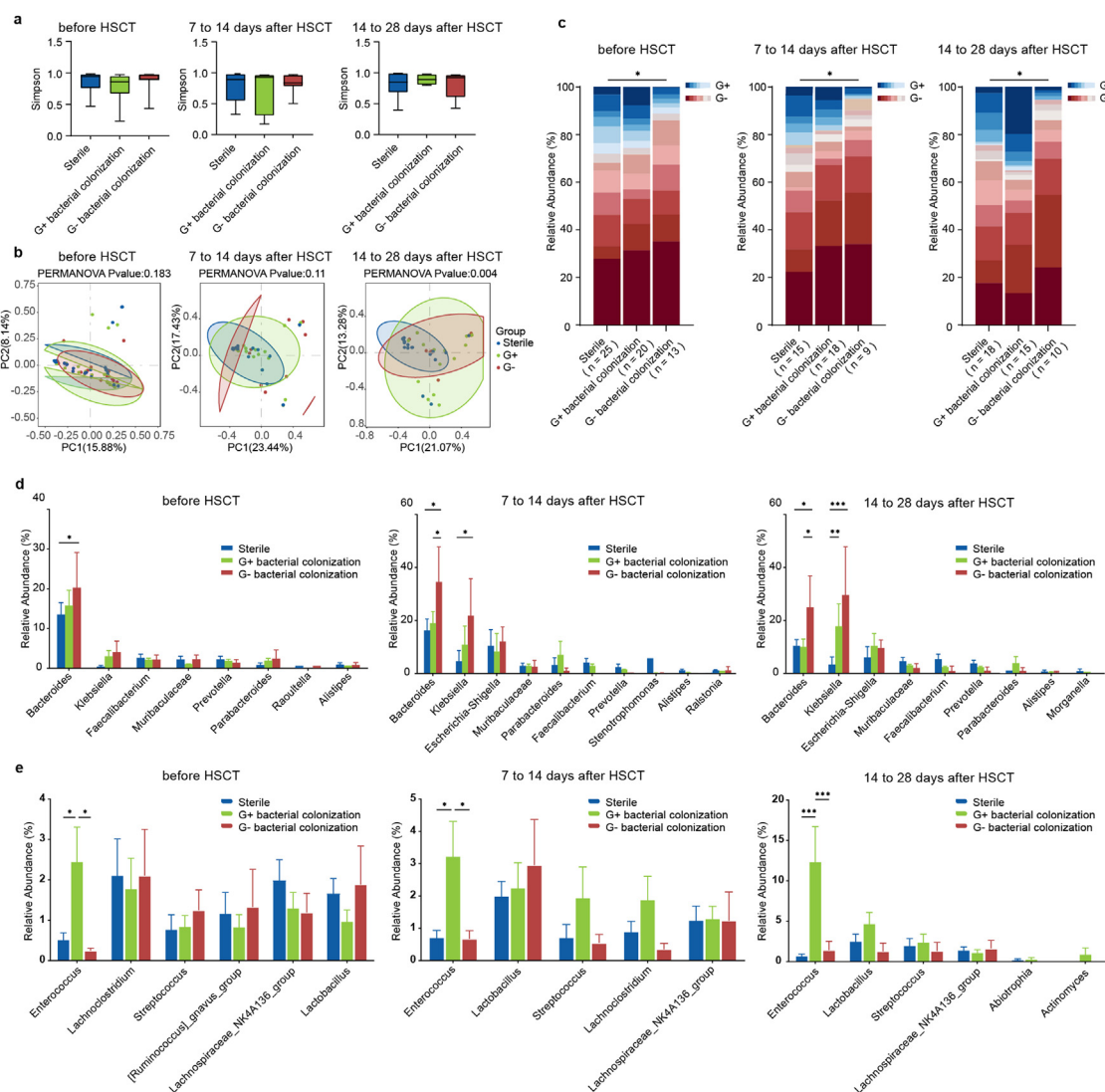


Figure 1. Differences in microbiota diversity and abundance among the sterile group, G^+ group, and G^- group before and after transplantation. Stool samples were collected before allo-HSCT and at 7 to 14 days and 14 to 28 days after allo-HSCT for 16S rRNA gene sequencing. (A) Simpson index of α -diversity in the 3 groups. (B) PCoA analysis of β -diversity in the 3 groups. (C) Relative abundances of gram-negative and gram-positive bacteria in stool samples. (D, E) Relative abundances of the top 15 microbial genera, including gram-negative bacteria (D) and gram-positive bacteria (E). * $P < .05$; ** $P < .01$; *** $P < .001$.

14 days after allo-HSCT, 88.1% versus 72% [$P = .049$]; at 14 to 28 days after allo-HSCT, 92.5% versus 73.9% [$P = .046$] (Figure 1C). The relative abundance of gram-positive or gram-negative bacteria was not significantly different between the sterile group and G^+ group before or after allo-HSCT (Figure 1C).

We next analyzed the top 15 microbial genera identified. In the genera of gram-negative bacteria, the relative abundance of *Bacteroides* spp was higher in the G^- group compared with the sterile group both before and after allo-HSCT (before HSCT, 20.50% versus 13.73% in the sterile group [$P = .037$]; at 7 to 14 days after allo-HSCT, 34.87% versus 16.60% in the sterile group [$P = .015$] and versus 19.26% in the G^+ group [$P = .045$]; at 14 to 28 days, 25.35% versus 10.70% in the sterile group [$P = .033$] and versus 10.36% in the G^+ group [$P = .031$]) (Figure 1D). The relative abundance percentage of *Klebsiella* spp also was higher in the G^- group compared with the sterile group after allo-HSCT (at 7 to 14 days, 22.09% versus 4.95% [$P = .024$]; at 14 to 28 days, 29.91% versus 3.71% [$P < .001$]) (Figure 1D). In the genera of gram-positive bacteria, the

relative abundance of *Enterococcus* spp was higher in the G^+ group compared with the other 2 groups both before and after allo-HSCT (before HSCT, 2.45% versus 1.04% in the sterile group [$P = .034$] and versus .24% in the G^- group [$P = .049$]; at 7 to 14 days after allo-HSCT, 3.23% versus .72% in the sterile group [$P = .023$] and versus .67% in the G^- group [$P = .041$]; at 14 to 28 days, 12.4% versus .71% in the sterile group [$P < .001$] and versus 1.41% in the G^- group [$P < .001$]) (Figure 1E). The 16S rRNA sequencing results showed differences in the abundances of specified genera to which the cultured microorganisms belong, such as *Klebsiella* spp and *Enterococcus* spp. The results suggest that the anal swab test results were consistent with the gram-negative or gram-positive bacteria abundance in the 16S rRNA sequencing results.

Immune Characteristics

We compared immune markers and plasma cytokines among the 3 groups after transplantation. The percentage of total lymphocytes was significantly higher in the G^- group compared with the sterile group at 2 months and 3 months

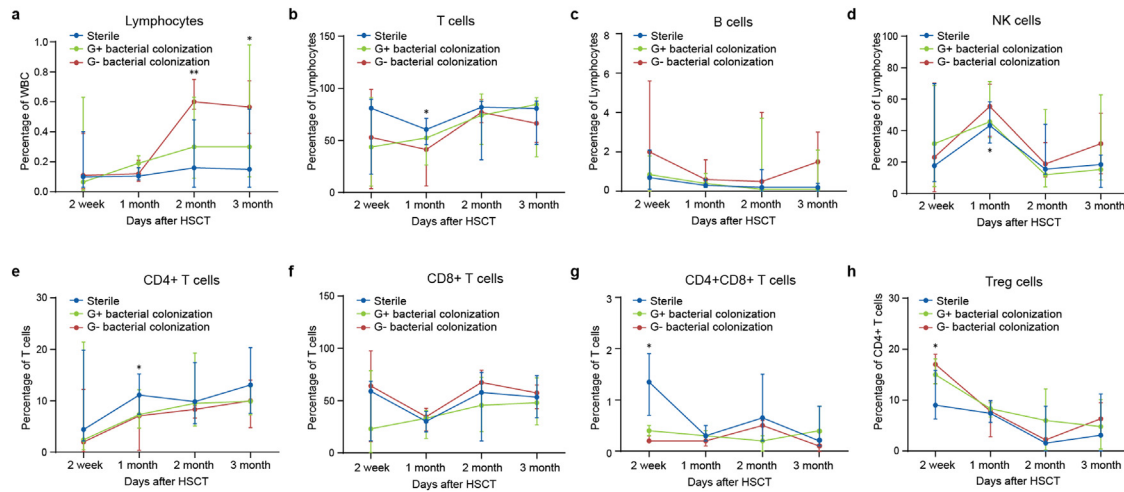


Figure 2. Immune cell reconstitution after transplantation. Peripheral blood samples were collected at 2 weeks, 1 month, 2 months, and 3 months after transplantation and processed for flow cytometry analysis. (A) Percentage of lymphocytes (CD45⁺). (B–D) Percentages of NK (CD3⁺CD16⁺CD56⁺) and T and B cells in lymphocytes. (E–G) Percentages of CD4⁺ T cells, CD8⁺ T cells, and CD4⁺CD8⁺ T cells in T cells. (H) Percentage of Tregs (CD25⁺CD127⁺) in CD4⁺ T cells. * $P < .05$; ** $P < .01$.

after transplantation (Figure 2A). Compared to the sterile group patients, G⁻ group patients had a higher percentage of NK cells (Figure 2B) and a lower percentage of T cells (Figure 2C) at 1 month after transplantation. With regard to T cell subsets, percentages of CD4⁺ T cells (Figure 2E) and CD4⁺CD8⁺ T (double-positive T; DPT) cells (Figure 2G) were lower in the G⁻ group than in the sterile group at 1 month and 2 weeks after transplantation, respectively. At 2 weeks post-transplantation, the percentage of regulatory T cells (Tregs) was higher in the G⁻ group than in the sterile group (Figure 2H). There was no significant difference in B cells and CD8⁺ T cells among the 3 groups (Figure 2D, F); however, levels of plasma TNF- α , IFN- γ , IL-6, and IL-17A were lower in the G⁻ group compared with the sterile group at 2 weeks after transplantation (Figure 3A–D), with no difference in IL-10 and IL-4 levels (Figure 3E–F).

aGVHD

The cumulative incidence of aGVHD was not significantly different among the 3 groups (Figure 4A). There was a trend toward a lower cumulative incidence of grade II–IV aGVHD in the G⁻ group compared with the sterile group (27.9% versus 46.6%; $P = .071$) (Figure 4B). Moreover, univariate analysis of the cumulative incidence of grade III–IV aGVHD showed a lower incidence rate in the G⁻ group compared with the sterile group (15.8% versus 36.1%; $P = .028$) (Figure 4C). Multivariate analysis identified anal swab gram-negative bacterial colonization as a protective factor against grade III–IV aGVHD (HR, .55; 95% CI, .317 to .96; $P = .035$) (Table 2).

At 7 to 14 days and 14 to 28 days after transplantation, the relative abundance of gram-negative bacteria in the stool of patients with grade 0–II aGVHD was higher than that in the stool of patients with grade III–IV aGVHD (days 7 to 14, 87.3%

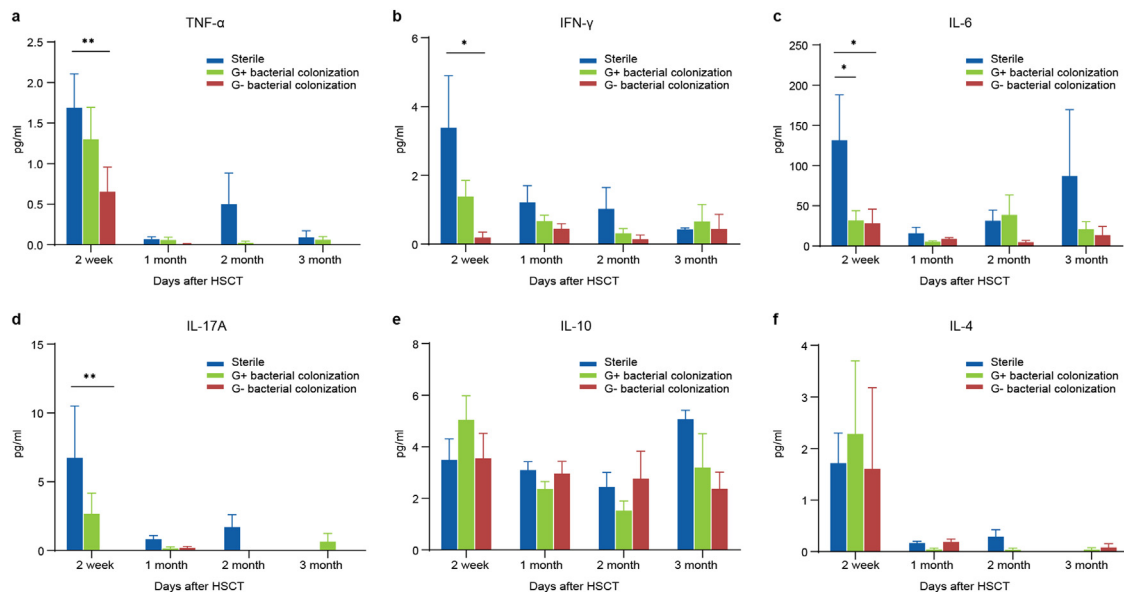


Figure 3. Levels of plasma cytokines after transplantation. Plasma samples were isolated from peripheral blood collected in EDTA anticoagulant tubes at 2 weeks, 1 month, 2 months, and 3 months after transplantation and processed for flow cytometry analysis. Repeated measures from the same participants with immune markers were not taken into account in these comparisons. Levels of plasma TNF- α , IFN- γ , IL-6, IL-17A, IL-10, and IL-4 were determined. * $P < .05$; ** $P < .01$.

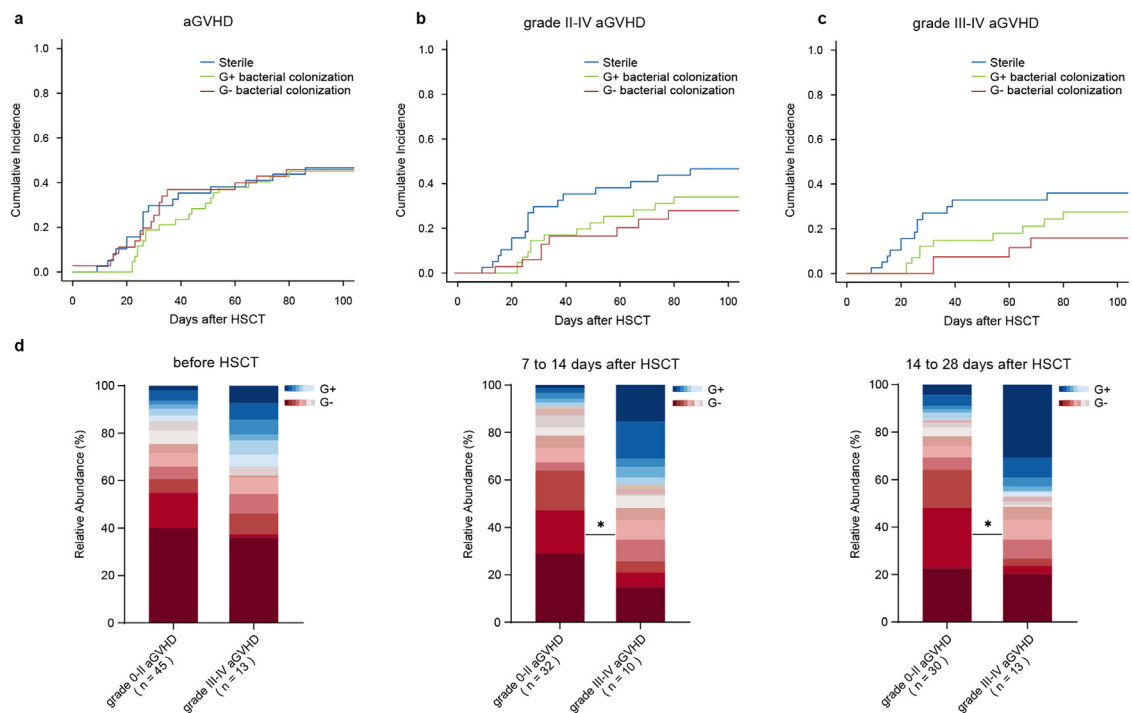


Figure 4. aGVHD and fecal bacteria characteristics after HSCT. (A) Cumulative incidence of aGVHD according to anal swab test results. (B) Cumulative incidence of grade II-IV aGVHD. (C) Cumulative incidence of grade III-IV aGVHD. (D) Relative abundances of gram-negative and gram-positive bacteria in stool samples of grade 0-II and III-IV aGVHD patients. * $P < .05$.

Table 2

Univariate and Multivariate Analysis of Outcomes Based on Patient and Transplant Characteristics

Variable	Univariate Analysis		Multivariate Analysis	
	HR (95% CI)	P Value	HR (95% CI)	P Value
Cumulative incidence of grade III-IV aGVHD				
Recipient age, ≤ 40 yr vs > 40 yr	1.493 (.693–3.217)	.307		
Recipient sex, female vs male	1.212 (.562–2.613)	.624		
Donor type, mismatched vs matched	1.756 (.709–4.351)	.224		
Conditioning (RIC vs MAC)	1.347 (.625–2.903)	.447		
Stem cell source				
BM vs PB	1.471 (.195–11.093)	.708		
BM vs PB + BM	1.519 (.194–11.875)	.69		
PB vs PB + BM	1.033 (.468–2.278)	.936		
Antithymocyte globulin, no vs yes	1.347 (.625–2.903)	.447		
Mononuclear cells, $\leq$ median vs $>$ median	1.464 (.685–3.129)	.325		
CD34 ⁺ cells, $\leq$ median vs $>$ median	.575 (.258–1.279)	.175	.504 (.223–1.140)	.100
Group				
Sterile vs G ⁺ bacterial colonization	.624 (.273–1.422)	.233	.599 (.270–1.325)	.210
Sterile vs G ⁻ bacterial colonization	.307 (.1–.941)	.028	.552 (.317–.960)	.035
G ⁺ bacterial colonization vs G ⁻ bacterial colonization	.492 (.154–1.568)	.244	.416 (.124–1.390)	.150
GRFS				
Recipient age, ≤ 40 yr vs > 40	1.159 (.632–2.125)	.633		
Recipient sex, female vs male	.832 (.444–1.559)	.566		
Donor type, mismatched vs matched	2.038 (.999–4.155)	.05	1.882 (.918–3.855)	.084
Conditioning, RIC vs MAC	1.428 (.775–2.632)	.253		
Stem cell source				
BM vs PB	2.975 (.404–21.924)	.285		
BM vs PB+BM	2.726 (.36–20.649)	.332		
PB vs PB+BM	.916 (.487–1.723)	.786		
Antithymocyte globulin, no vs yes	1.52 (.819–2.819)	.184		
Mononuclear cells, $\leq$ median vs $>$ median	.864 (.47–1.588)	.638		
CD34 ⁺ cells, $\leq$ median vs $>$ median	.566 (.304–1.053)	.073	.541 (.289–1.013)	.055
Group				
Sterile vs G ⁺ bacterial colonization	.72 (.367–1.415)	.341	.811 (.411–1.599)	.545
Sterile vs G ⁻ bacterial colonization	.401 (.178–.903)	.027	.416 (.183–.947)	.037
G ⁺ bacterial colonization vs G ⁻ bacterial colonization	.556 (.248–1.249)	.155	.514 (.227–1.161)	.109

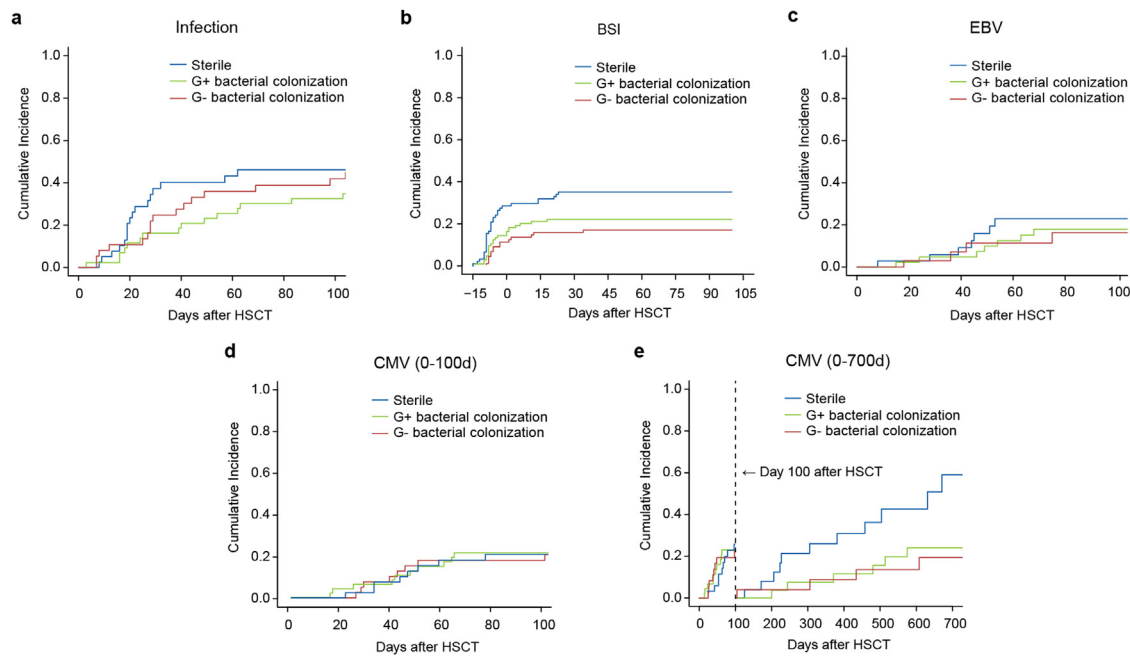


Figure 5. Infectious complications after allo-HSCT. (A) Cumulative incidence of infection within 100 days after transplantation according to anal swab test results. (B) Cumulative incidence of BSI within 100 days after transplantation. (C) Cumulative incidence of EBV reactivation within 100 days after transplantation. (D) Cumulative incidence of CMV reactivation within 100 days after transplantation. (E) Cumulative incidence of CMV reactivation within 2 years after transplantation. The horizontal dotted line on day 100 is a landmark analysis for the subset of patients who had not yet exhibited CMV reactivation by day 100.

versus 57.8% [$P = .043$]; days 14 to 28, 85.2% versus 52.8% [$P = .046$]) (Figure 4D).

Infectious Complications

The cumulative incidence of BSI within 100 days after transplantation was lower in the G⁻ group compared with the sterile group (17.2% versus 35.2%; $P = .009$) (Figure 5B). There were no significant differences among the sterile, G⁺, and G⁻ groups with regard to infection (46.2% versus 32.5% versus 41.8%; Figure 5A), EBV reactivation (22.9% versus 17.9% versus 16.3%; Figure 5C), and CMV reactivation (22.5% versus 23.1% versus 22.4%; Figure 5D) up to day 100 post-transplantation. However, the analysis of patients without CMV infection on day 100 post-transplantation showed a lower rate of CMV reactivation after 100 days to 2 years post-transplantation in the G⁻ group compared with the sterile group (13.6% versus 59%; $P = .023$; Figure 5E).

NRM and Survival

The median duration of follow-up was 14.48 months (range 5.7 to 33.72 months), and 18 patients died. The 2-year NRM was 10.8% in the sterile group, 9.9% in the G⁺ group, and 12.3% in the G⁻ group (Figure 6A). There were no statistically

significant differences in the cumulative incidence of NRM among the 3 groups, as well as no significant differences in 2-year OS (89.7% versus 84.1% versus 91.9%; Figure 6B) or DFS (89.7% versus 77.3% versus 86.5%; Figure 6C). Furthermore, univariate analysis revealed a better 2-year GRFS in the G⁻ group compared with the sterile group (75.7% versus 59%; $P = .027$) (Figure 6D). In multivariate analysis, gram-negative anal swab colonization was associated with better GRFS (HR, .04; 95% CI, .18 to .95; $P = .037$) (Table 2).

DISCUSSION

Intestinal microbiota dysbiosis is associated with poor clinical outcomes, including infection and aGVHD, in allo-HSCT recipients [11]. The methodology for evaluating the intestinal microbiota has evolved from traditional culture to modern molecular biology methods. Although the development of various high-throughput sequencing technologies has enriched our understanding of enterobacteria at various levels, depth bias in detection technologies remains unavoidable, and there are still advantages of traditional culture methods for evaluating less concentrated samples [12]. Cost and timeliness are other factors that must be considered when choosing a

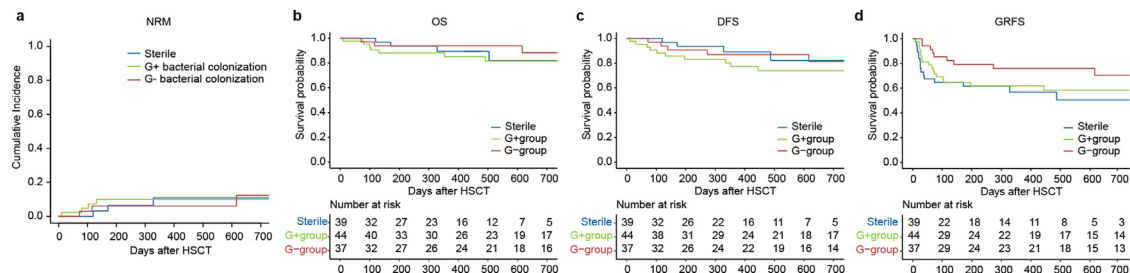


Figure 6. NRM and survival. (A) Cumulative incidence of NRM up to 2 years after allo-HSCT according to anal swab test results. (B-D) Kaplan-Meier estimates of OS, DFS, and GRFS up to 2 years after allo-HSCT.

detection method related to its clinical application, especially for intervention and guidance during allo-HSCT [13].

Some potential advantages of anal swab testing in clinical virus detection have been reported recently, demonstrating its applicability and reliability to a certain extent [14,15]. Anal swab testing is described as a bridging method for further testing of stool samples [16]; however, its use for clinical flora microbial detection has not attracted sufficient attention. In this study, anal swab testing before transplantation was used to detect intestinally colonized microorganisms in allo-HSCT recipients. Consistency between the anal swab test results and the gram-negative or gram-positive bacteria abundance of 16S rRNA sequencing results was verified, and relationships between anal swab test results and the clinical outcomes of allo-HSCT were evaluated.

According to fecal 16S rRNA gene sequencing results, the level of gram-negative bacteria was higher in the G⁻ group compared with the other groups both before and after transplantation, demonstrating the high confidence level in anal swab test results. Previous studies have found decreased microbial diversity and significant changes in intestinal microbial composition after HSCT [17,18]. Our analysis of microbial diversity in the 3 study groups revealed a significant difference in β -diversity at 14 to 28 days after transplantation but no differences in α -diversity both before and after transplantation. In the G⁻ group, the abundance of fecal gram-negative bacteria was increased, with more pronounced increases in *Bacteroides* spp and *Klebsiella* spp. Sofi et al. [19] reported that oral gavage of *Bacteroides fragilis* increased the diversity of intestinal microbiota, repaired the integrity of the intestinal barrier, and alleviated aGVHD and chronic GVHD in mice. Moreover, the *Bacteroides* genus has a negative correlation with the occurrence of severe aGVHD [20]. These findings indicate that an increased abundance of specific gram-negative bacteria is associated with improved clinical outcomes of transplantation. Therefore, detection of gram-negative bacterial colonization by anal swabs before transplantation may predict relevant clinical outcomes after transplantation.

Further analysis of the incidence of aGVHD in allo-HSCT recipients revealed that anal swab gram-negative bacterial colonization was associated with a reduced cumulative incidence of grade III-IV aGVHD within 100 days and improved GFRS at 2 years. The relative abundance of gram-negative bacteria was higher in patients with grade 0-II aGVHD compared with patients with grade III-IV aGVHD. These results indicate that the anal swab test may have predictive significance for the occurrence of severe aGVHD.

Because several studies have shown that the intestinal microbiota is critical in immune homeostasis [21–23], we examined immune cell reconstitution and cytokine levels after transplantation. Our results showed faster immune reconstitution of lymphocytes in the G⁻ group compared with the sterile group, owing mainly to higher levels of NK cells. The percentage of T cells was lower in the G⁻ group, with decreases in both CD4⁺ T cells and DPT cells. aGVHD is caused by alloreactive donor T cells, which are transferred into patients during HSCT [24]. DPT cells, developed from antigen-stimulated CD8⁺ T cells, might mediate xeno-GVHD pathology and predict the incidence of grade $\geq$ II GVHD [25]. Tregs can inhibit the proliferation and activation of donor-derived effector T cells (Teffs), enhance the immune tolerance of Teffs to patient tissues, and reduce the severity of aGVHD. Our results showed a higher percentage of Tregs in the G⁻ group compared with the sterile group at 2 weeks post-transplantation; however, the

percentage of T lymphocytes was extremely low in many patients at 2 weeks after HSCT. Confirmation of the data for T cell subsets awaits a larger number of patient samples for analysis. In addition, plasma concentrations of the proinflammatory cytokines TNF- α , IFN- γ , IL-6, and IL-17A were lower in the G⁻ group compared with the sterile group at 2 weeks after transplantation, which may be related to the low risk of severe aGVHD. Overall, the results of anal swab testing before transplantation may be associated with early immune homeostasis after transplantation.

The intestinal microbiota also can influence infection by modulating the host immune response [26]. Different components of the intestinal microbiota act in a direct or indirect manner to promote or suppress viral infection. Conversely, the viruses also affect intestinal microbiota diversity and composition [27]. Our results show an association between anal swab colonization with gram-negative bacteria and lower incidences of BSI within 100 days post-transplantation and CMV reactivation at 100 days to 2 years post-transplantation. Overall, our results suggest that gram-negative bacterial colonization may improve clinical outcomes by affecting the immune response.

In general, our data indicate a high confidence in microbial detection with anal swab testing. In our analysis of the association between pretransplantation anal swab colonization and clinical outcomes after allo-HSCT, the anal swab test results were consistent with the gram-negative or positive bacteria abundance of 16S rRNA sequencing results. Moreover, gram-negative bacterial colonization before allo-HSCT affected the immune response and inflammatory cytokine production. Gram-negative bacterial colonization was associated with a lower risk of grade III-IV aGVHD and lower incidences of BSI and CMV reactivation. In view of the accuracy and predictive value of anal swabs in this study, more attention to and exploration of the advantages of clinical anal swab testing for diagnosis and treatment of blood diseases is warranted.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jtct.2023.07.018.

REFERENCES

1. Bazinet A, Popradi G. A general practitioner's guide to hematopoietic stem-cell transplantation. *Curr Oncol*. 2019;26:187–191. <https://doi.org/10.3747/co.26.5033>.
2. Lin D, Hu B, Li P, Zhao Y, Xu Y, Wu D. Roles of the intestinal microbiota and microbial metabolites in acute GVHD. *Exp Hematol Oncol*. 2021;10:49. <https://doi.org/10.1186/s40164-021-00240-3>.
3. Galloway-Peña J, Brumlow C, Shelburne S. Impact of the microbiota on bacterial infections during cancer treatment. *Trends Microbiol*. 2017;25:992–1004. <https://doi.org/10.1016/j.tim.2017.06.006>.
4. Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med*. 2020;382:822–834. <https://doi.org/10.1056/NEJMoa1900623>.
5. Peled JU, Devlin SM, Staffas A, et al. Intestinal microbiota and relapse after hematopoietic-cell transplantation. *J Clin Oncol*. 2017;35:1650–1659. <https://doi.org/10.1200/JCO.2016.70.3348>.
6. Miao Q, Ma Y, Wang Q, et al. Microbiological diagnostic performance of metagenomic next-generation sequencing when applied to clinical practice. *Clin Infect Dis*. 2018;67(suppl_2):S231–S240. <https://doi.org/10.1093/cid/ciy693>.
7. Blazar BR, Hill GR, Murphy WJ. Dissecting the biology of allogeneic HSCT to enhance the GvT effect whilst minimizing GVHD. *Nat Rev Clin Oncol*. 2020;17:475–492. <https://doi.org/10.1038/s41571-020-0356-4>.
8. Timler M, Timler W, Bednarsz A, et al. Identification and preliminary hierarchisation of selected risk factors for carbapenemase-producing Enterobacteriaceae (CPE) colonisation: a prospective study. *Int J Environ Res Public Health*. 2023;20:1960. <https://doi.org/10.3390/ijerph20031960>.
9. Przepiorka D, Weisdorf D, Martin P, et al. 1994 Consensus Conference on Acute GVHD Grading. *Bone Marrow Transplant*. 1995;15:825–828.
10. US Department of Health and Human Services. *Common Terminology Criteria for Adverse Events (CTCAE) version 5.0*. Bethesda, MD: National Institutes of Health; 2017.
11. Rey-Mariño A, Francino MP. Nutrition, gut microbiota, and allergy development in infants. *Nutrients*. 2022;14:4316. <https://doi.org/10.3390/nu14204316>.
12. Hugon P, Lagier JC, Colson P, Bittar F, Raoult D. Repertoire of human gut microbes. *Microb Pathog*. 2017;106:103–112. <https://doi.org/10.1016/j.micpath.2016.06.020>.
13. Granroth G, Khera N, Arana, Yi C. Progress and challenges in survivorship after acute myeloid leukemia in adults. *Curr Hematol Malig Rep*. 2022;17:243–253. <https://doi.org/10.1007/s11899-022-00680-6>.
14. Isoldi S, Mallardo S, Marcellino A, et al. The comprehensive clinic, laboratory, and instrumental evaluation of children with COVID-19: a 6-months prospective study. *J Med Virol*. 2021;93:3122–3132. <https://doi.org/10.1002/jmv.26871>.
15. Zhong H, Wang Y, Shi Z, et al. Characterization of respiratory microbial dysbiosis in hospitalized COVID-19 patients. *Cell Discov*. 2021;7:23. <https://doi.org/10.1038/s41421-021-00257-2>.
16. Coryell MP, Iakiviak M, Pereira N, et al. A method for detection of SARS-CoV-2 RNA in healthy human stool: a validation study. *Lancet Microbe*. 2021;2:e259–e266. [https://doi.org/10.1016/S2666-5247\(21\)00059-8](https://doi.org/10.1016/S2666-5247(21)00059-8).
17. Jorgensen M, Norgaard JC, Ilett EE, et al. Metabolic potential of the gut microbiome is significantly impacted by conditioning regimen in allogeneic hematopoietic stem cell transplantation recipients. *Int J Mol Sci*. 2022;23:11115. <https://doi.org/10.3390/ijms231911115>.
18. Upadhyay Banskota S, Skupa SA, El-Gamal D, D'Angelo CR. Defining the role of the gut microbiome in the pathogenesis and treatment of lymphoid malignancies. *Int J Mol Sci*. 2023;24:2309. <https://doi.org/10.3390/ijms24032309>.
19. Sofi MH, Wu Y, Ticer T, et al. A single strain of *Bacteroides fragilis* protects gut integrity and reduces GVHD. *JCI Insight*. 2021;6:e136841. <https://doi.org/10.1172/jci.insight.136841>.
20. Chong PP, Koh AY. The gut microbiota in transplant patients. *Blood Rev*. 2020;39:100614. <https://doi.org/10.1016/j.blre.2019.100614>.
21. Schlechte J, Zucoloto AZ, Yu IL, et al. Dysbiosis of a microbiota-immune metasystem in critical illness is associated with nosocomial infections. *Nat Med*. 2023;29:1017–1027. <https://doi.org/10.1038/s41591-023-02243-5>.
22. Miltiadous O, Waters NR, Andrlóvá H, et al. Early intestinal microbial features are associated with CD4 T-cell recovery after allogeneic hematopoietic transplant. *Blood*. 2022;139:2758–2769. <https://doi.org/10.1182/blood.2021014255>.
23. van der Lelie D, Oka A, Taghavi S, et al. Rationally designed bacterial consortia to treat chronic immune-mediated colitis and restore intestinal homeostasis. *Nat Commun*. 2021;12:3105. <https://doi.org/10.1038/s41467-021-23460-x>.
24. Holtan SG, Pasquini M, Weisdorf DJ. Acute graft-versus-host disease: a bench-to-bedside update. *Blood*. 2014;124:363–373. <https://doi.org/10.1182/blood-2014-01-514786>.
25. Hess NJ, Turicek DP, Riendeau J, et al. Inflammatory CD4/CD8 double-positive human T cells arise from reactive CD8 T cells and are sufficient to mediate GVHD pathology. *Sci Adv*. 2023;9:eadf0567. <https://doi.org/10.1126/sciadv.adf0567>.
26. Bassetti M, Bandera A, Gori A. Therapeutic potential of the gut microbiota in the management of sepsis. *Crit Care*. 2020;24:105. <https://doi.org/10.1186/s13054-020-2780-3>.
27. Yang M, Yang Y, He Q, et al. Intestinal microbiota—a promising target for antiviral therapy? *Front Immunol*. 2021;12:676232. <https://doi.org/10.3389/fimmu.2021.676232>.